Sovereign Process Mining:

Algebraic Derivations and Differential Calculus of Cryptographic Execution Spaces

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Abstract

As Process-Aware Information Systems (PAIS) decentralize into autonomic networks, classical process mining algorithms must evolve from a-posteriori statistical heuristics into rigorous, executable mathematical authorities. In this paper, we present the absolute algebraic and calculus-derived foundations of `wasm4pm`, a sovereign WebAssembly ecosystem. By deriving process discovery, token-based replay, and optimal A* alignments from first mathematical principles, we eliminate empirical ambiguity. We formulate event logs as continuous measure spaces, map Partially Ordered Workflow Languages (POWL) via graph-theoretic tensor projections, and derive the exact Linear Programming (LP) relaxations required for $O(1)$ conformance heuristics. The culmination is a mathematically impenetrable framework where every topological state transition binds uniquely to a cryptographic BLAKE3 receipt, establishing zero-trust process execution.

1 Introduction and Topological Foundations

Classical process mining operates on the assumption of a static, retrospective event log residing in a trusted central repository. In decentralized, trustless environments, this assumption collapses. To elevate process mining to an executable authority, we must reconstruct its foundations using rigorous algebra and calculus.

Let $\mathcal{U}$ be the universe of all observable system states. We define an *event* not merely as a tuple, but as an instantaneous state transition in a continuous time domain.

Definition 1 (Continuous Event Space)**.** *Let $\mathcal{T} \subset \mathbb{R}_{\geq 0}$ represent continuous time. The event space $\mathcal{E}$ is a manifold mapping time to state differentials:*

$$\mathbf{e}(t) = \frac{\partial \mathbf{S}}{\partial t}$$

where $\mathbf{S}(t) \in \mathcal{U}$ is the system state at time t. An observed event e_i is the integral over a localized pulse (e.g., a Dirac delta function $\delta(t - t_i)$):

$$e_i = \int_{t_i-\epsilon}^{t_i+\epsilon} \frac{\partial \mathbf{S}}{\partial t}\, dt \approx \Delta \mathbf{S}_i$$

A trace σ is a trajectory through this state space, parameterized by a case identifier $c \in \mathcal{C}$. An event log L is a probability measure space $(\Omega, \mathcal{F}, \mathbb{P})$ over the set of all possible trajectories.

2 The Calculus of Conformance Checking

Conformance checking seeks to measure the distance between the observed trajectory $\sigma \in L$ and the language $L(M)$ generated by a formal process model M.

2.1 Petri Nets and the Incidence Matrix

We represent a Petri net algebraically. Let P be a set of places ($|P| = m$) and T be a set of transitions ($|T| = n$). The structure is completely defined by the Incidence Matrix $\mathbf{W} \in \mathbb{Z}^{m\times n}$.

$$\mathbf{W}_{i,j} = \mathbf{W}^+_{i,j} - \mathbf{W}^-_{i,j}$$

where $\mathbf{W}^+$ is the post-set matrix and $\mathbf{W}^-$ is the pre-set matrix. The state equation governing token dynamics from marking $\mathbf{M}_k$ to $\mathbf{M}_{k+1}$ via transition vector $\mathbf{u}_k$ is:

$$\mathbf{M}_{k+1} = \mathbf{M}_k + \mathbf{W}\mathbf{u}_k$$

2.2 Optimal Alignments via Variational Calculus

An alignment γ maps a trace σ to a model trajectory σ_M. Let $\mathbf{x} \in \{0,1\}^{|\sigma|}$ be an indicator vector for synchronous moves, $\mathbf{y}$ for log moves, and $\mathbf{z}$ for model moves. We define a continuous cost functional $J(\gamma)$ over the alignment path $\Gamma(s)$:

$$J(\gamma) = \int_\Gamma \left(c_L \|\dot{\mathbf{y}}(s)\|^2 + c_M \|\dot{\mathbf{z}}(s)\|^2 \right) ds$$

Minimizing this cost function is equivalent to solving the Euler-Lagrange equations. In the discrete domain, this is solved via the A* algorithm.

2.3 Derivation of the State Equation Heuristic

To guarantee admissibility for A*, the heuristic function $h(\mathbf{M})$ must satisfy $h(\mathbf{M}) \leq h^*(\mathbf{M})$. We derive $h(\mathbf{M})$ by relaxing the integer constraint on the firing vector $\mathbf{x}$ to the continuous domain $\mathbb{R}^n_{\geq 0}$. We formulate the Linear Program (LP):

$$\begin{aligned} \min \quad & \mathbf{c}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{M}_{final} - \mathbf{M} = \mathbf{W}\mathbf{x} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$$

The dual of this LP provides the tightest lower bound. By computing the Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathbf{c}^T\mathbf{x} + \boldsymbol{\lambda}^T(\mathbf{M}_{final} - \mathbf{M} - \mathbf{W}\mathbf{x})$, and setting the gradient to zero:

$$\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{c} - \mathbf{W}^T \boldsymbol{\lambda} = \mathbf{0} \implies \mathbf{W}^T \boldsymbol{\lambda} = \mathbf{c}$$

The maximal lower bound is $\max \boldsymbol{\lambda}^T(\mathbf{M}_{final} - \mathbf{M})$. This exact algebraic derivation ensures zero information loss during conformance execution within the `wasm4pm` engine.

3 Process Discovery: Differential Gradients on the Directly-Follows Graph

Process discovery algorithms like the Inductive Miner rely on the Directly-Follows Graph (DFG). Let $\mathcal{A}$ be the set of activities. The DFG is an adjacency matrix $\mathbf{D} \in \mathbb{N}^{|\mathcal{A}| \times |\mathcal{A}|}$.

To filter noise, we treat the thresholding problem as continuous optimization. We define an activation function $\sigma_{\alpha,\beta}(x)$ representing the edge retention probability:

$$\sigma_{\alpha,\beta}(x) = \frac{1}{1 + e^{-\alpha(x-\beta)}}$$

The discovery objective is to maximize the likelihood of the model given the data, regularized by structural complexity (Occam's Razor). The gradient descent over the edge weights $\mathbf{D}$ is:

$$\nabla_{\mathbf{D}} \mathcal{J} = \frac{\partial}{\partial \mathbf{D}} \left[\sum_{\sigma \in L} \log \mathbb{P}(\sigma \mid \mathbf{D}) - \lambda \|\mathbf{D}\|_1 \right]$$

By applying rigorous graph cuts recursively over this optimized tensor, we map the process into a Partially Ordered Workflow Language (POWL).

4 POWL to Petri Net Projections

Let $\mathcal{G}$ be a non-block-structured Choice Graph in POWL. We project $\mathcal{G} \to PN$ algebraically. The mapping operator Π preserves language monotonically: $L(\mathcal{G}) \subseteq L(\Pi(\mathcal{G}))$. For irreducible subnets, we emit a Single-Entry Single-Exit (SESE) silent τ-transition routing block to maintain the marking invariant $\sum_{p \in P} M(p) = 1$. The invariant is proven by verifying that the null space of $\mathbf{W}^T$ contains the vector of all ones: $\mathbf{W}^T \mathbf{1} = \mathbf{0}$.

5 Cryptographic Binding and Closure

The `wasm4pm` execution authority requires that every mathematical transformation $\mathcal{F} : S_i \to S_{i+1}$ yields a cryptographic proof.

$$H_{i+1} = \text{BLAKE3}(H_i \parallel \mathcal{F} \parallel S_{i+1})$$

This unforgeable recursive digest structurally guarantees that the entire calculus derived in this paper was executed exactly as defined, immune to adversarial tampering.

6 Conclusion

By reconstructing process mining from fundamental calculus, linear algebra, and topology, we elevate it from empirical estimation to mathematical certainty. The sovereign `wasm4pm` environment ensures that these algebraic derivations are executed immutably at the edge.